

Evaluating Hybrid Semantic–Lexical Retrieval Using Voyage-law-2 and SPLADE for EU Climate Legal Documents

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Abstract—Legal Information Retrieval in Statutory Corpora poses specific challenges because of the use of formulaic language, hierarchical structures of authority, and the need for doctrinal precision. Previous work has shown that there is a fundamental trade-off between term-matching-based lexical models and dense semantic embedding-based models in legal tasks. Lexical models such as BM25 are still very effective baselines in highly structured texts, while dense models are better at semantic abstraction but may be less doctrinally sound. Recent work on hybrid models that combine both lexical and semantic information has shown systematic gains in regulatory and multilingual legal tasks. However, previous work is still quite scattered across datasets, and there is a lack of comprehensive evaluation of retrieval effectiveness, reliability, and computational efficiency simultaneously in statute-level corpora. This paper explores whether a hybrid retrieval framework that combines Voyage-law-2 semantic embeddings with SPLADE sparse lexical expansion outperforms BM25 and dense-only baselines in EU climate law documents. By linking this work to previous findings on lexical-dense trade-offs, document-level retrieval mismatch, and multi-stage architectures for legal information retrieval, this paper argues that hybrid sparse-dense retrieval is a principled and testable approach for high-stakes legal reasoning tasks.

Index Terms—Legal Information Retrieval, Hybrid Retrieval, SPLADE, Voyage-law-2, Dense Retrieval, Sparse Retrieval, EU Climate Law, Retrieval-Augmented Generation

I. INTRODUCTION

Legal information retrieval is a paradigm that differs significantly from open-domain search because the meaning of the law is contingent on exact wording, cross-references, and authority relationships. This is particularly important in the context of EU climate and environmental law, where highly formalized and interconnected texts imply that retrieval errors may result in the misinterpretation of doctrine rather than semantic approximation.

Recent work has identified a tension between lexical and dense retrieval paradigms in legal information retrieval. Lexical models such as BM25 are robust in formulaic corpora where exact wording overlap is indicative of relevance [1], while dense retrievers are more adept at modeling semantic abstraction in reasoning tasks [8]. However, dense models also introduce new failure cases such as Document-Level Retrieval

Mismatch (DRM), where semantically similar passages from globally irrelevant documents are retrieved based on shared boilerplate [12]. Empirical studies also demonstrate that off-the-shelf dense retrievers are less effective than lexical baselines on specialized legal datasets without domain adaptation [14].

Although the effectiveness of hybrid sparse-dense retrieval has been shown to improve legal search, its effectiveness has not yet been systematically assessed for the level of statute-level EU climate law. Current research either investigates general legal corpora or computationally intensive multi-stage refinement pipelines, which raises questions about whether a well-designed first-stage hybrid retriever alone can be effective in highly formalized regulatory texts.

This practicum evaluates a hybrid retrieval system combining Voyage-law-2 semantic embeddings with SPLADE lexical expansion and compares it with lexical (BM25) and dense-only baselines on EU climate-law documents to determine whether sparse–dense neural fusion offers a principled and practically viable improvement for statute-level legal reasoning.

II. THE TENSION BETWEEN LEXICAL AND DENSE RETRIEVAL IN LEGAL INFORMATION RETRIEVAL

Recent studies illustrate the existence of a trade-off between lexical term matching models and dense embedding-based models for legal information retrieval. Mori et al. [1] illustrate that BM25 is still a competitive baseline for formulaic legal texts, especially when the statutory language is repetitive and structurally rigid. Dense models tend to perform suboptimally in such scenarios because semantic similarity either ignores legally binding phrasing or gets sidetracked by noise in longer queries.

However, Zheng et al. [8] illustrate that lexical models tend to deteriorate in reasoning-intensive scenarios where relevant documents have little direct term overlap with the query. In such cases, semantic models are more adept at analogical reasoning. Reuter et al. [12] further identify the existence of a Document-Level Retrieval Mismatch (DRM) problem, where dense retrievers tend to pick passages from

the wrong documents because of shared structural boilerplate. The LRAGE evaluation framework [14] confirms the existence of such a problem, where off-the-shelf dense retrievers tend to perform suboptimally compared to BM25 in legal information retrieval tasks without domain-specific adaptation.

These studies thus confirm that neither lexical nor dense models are universally superior in legal information retrieval tasks. Lexical models are superior in preserving statutory precision, whereas dense models are superior in semantic abstraction. Owing to the highly structured nature of documents related to climate laws in the EU, dense-only systems are likely to retrieve semantically similar but incorrect passages, and lexical-only systems are likely to face issues of vocabulary mismatch. This study aims to determine if the combination of Voyage-law-2 embeddings and SPLADE expansion can fill this gap.

III. SPARSE NEURAL AND HYBRID LEXICAL-SEMANTIC RETRIEVAL SOLUTIONS

Hybrid retrieval systems combine lexical matching with neural semantic modeling, balancing precision with flexibility. SPLADE, originally proposed by Formal et al. [5] and later made more efficient by Lassance and Clinchant [4], generates sparse term expansions drawn from the BERT vocabulary while remaining fully compatible with inverted indexes. It addresses vocabulary mismatch issues while maintaining efficiency and interpretability.

Empirical research also confirms the complementarity of hybrid retrieval. Rayo et al. [3] show that a combination of BM25 with fine-tuned sentence transformers leads to a substantial boost in Recall@10 and MAP@10 on regulatory corpora. LegalRAG [10] illustrates that hybrid retrieval systems outperform single-mode systems for multilingual legal texts, especially in low-resource environments. Ho et al. [13] introduce structural signals like authority ranking and citation weighting, concluding that semantic similarity alone is not enough for doctrinal accuracy.

On the whole, the lexical and semantic cues seem to be complementary rather than substitutive. Contrary to the previous systems that combined BM25 with general dense encoders, this practicum will assess the hybrid model consisting of two specialized neural modules: Voyage-law-2 embeddings and SPLADE expansion. The sparse-dense neural fusion is designed for statute-level retrieval of EU climate law and aims to determine whether hybridization can mitigate the trade-off between exact term matching and semantic reasoning.

IV. MULTI-STAGE AND ADAPTIVE RETRIEVAL REFINEMENT IN LEGAL RAG SYSTEMS

Legal retrieval systems are increasingly using multi-stage and adaptive refinement techniques. HyPA-RAG [2] brings in adaptive parameter adjustment techniques, which vary depending on the complexity of the question. LexRAG [7] also shows that multi-turn legal conversations need refinement techniques to deal with topic drifts and implicit legal semantics.

The NOWJ@COLIEE challenge setting [11] presents a multi-stage architecture consisting of an initial recall-focused search, followed by semantic reranking and LLM-based entailment verification, which performs well on legal case retrieval tasks. LegalRAG [10] also adds secondary relevance validation layers to remove irrelevant passages before generation.

These experiments indicate that single-stage dense retrieval alone is not sufficient for high-stakes legal reasoning tasks. However, the computational cost of multi-stage systems is quite high. The current study fills an important empirical gap: whether a strong first-stage hybrid retriever (Voyage-law-2 + SPLADE) can provide reliable statute-level retrieval without necessarily needing complex adaptive or LLM-rich refinement layers. This study will examine whether architectural strength in the retrieval layer alone can mitigate the need for computationally expensive multi-stage systems.

V. RELIABILITY AND MISMATCH CHALLENGES

Reliability in legal RAG requires more than high recall; it demands correct source attribution, grounding coherence across dialogue turns, and terminological precision in generated outputs. Reuter et al. [12] formally quantify Document-Level Retrieval Mismatch (DRM), a failure mode in which retrievers select plausible chunks from structurally similar but incorrect documents. Evaluated on LegalBench-RAG across contracts (CUAD), merger agreements (MAUD), NDAs (ContractNLI), and privacy policies (PrivacyQA), standard dense pipelines exhibited DRM exceeding 95% on ContractNLI, where boilerplate homogeneity systematically confuses vector-similarity-based retrieval. Their proposed Summary-Augmented Chunking (SAC) prepends a concise LLM-generated document summary (150 characters) to each chunk before embedding, injecting global identity into otherwise isolated vectors. SAC substantially reduced DRM and improved both precision and recall across all datasets. Notably, generic summaries outperformed legally richer expert-guided ones, suggesting that broad semantic cues provide more robust query alignment than dense, structured legal language, a finding directly applicable to EU climate-law corpora, which share the repetitive cross-referencing structures that drive DRM in contract datasets.

Li et al. [7] address multi-turn grounding through LexRAG, the first RAG benchmark for legal consultation conversations, comprising 1,013 expert-annotated five-turn dialogues and 17,228 candidate legal articles. Legal consultations compound retrieval difficulty: queries involve pronoun resolution, topic shifts, and implicit legal intent that defeat keyword matching. Across BM25, BGE, GTE-Qwen2, and text-embedding-3 retrievers with four processing strategies, the best configuration achieved only 33.33% Recall@10. Critically, generation quality degraded when models received retriever-sourced context compared to gold articles, confirming that noisy retrieval propagates directly into unreliable outputs and that first-stage retrieval quality is the binding constraint on system reliability.

Chouhan and Gertz [9] add a terminological dimension through LexDrafter, which retrieves existing EU legislative

definitions from EUR-Lex to ground the drafting of new statutory terms, ensuring cross-document definitional coherence in EU regulatory corpora. Together, these three studies establish that legal RAG reliability is multi-dimensional, correct document attribution, conversational grounding, and terminological consistency must each be satisfied independently of one another.

VI. STRUCTURAL AND AUTHORITY MODELING

Legal relevance is governed not only by textual similarity but by the doctrinal hierarchy of sources. Wiratunga et al. [6] demonstrate through CBR-RAG that treating past cases as structured reasoning units and retrieving argumentatively analogous decisions improves factual alignment over standard dense retrieval, which treats all passages as interchangeable regardless of doctrinal status.

Ho et al. [13] extend this to statutory law via a knowledge-graph-based RAG system for copyright fair use, encoding court hierarchy and citation importance as explicit graph edges to prioritise authoritative over merely similar sources. For EU climate-law retrieval, where primary directives, implementing regulations, and delegated acts form a strict interpretive hierarchy, these findings highlight the risk of equal-weight retrieval surfacing legally subordinate material.

While the present study does not implement a knowledge graph, these findings motivate evaluating whether hybrid retrieval precision correlates with doctrinal relevance rather than surface-level semantic proximity alone.

VII. EVALUATION GAPS AND POSITIONING OF THIS STUDY

Standard metrics such as Recall@k and MRR are insufficient proxies for legal retrieval quality. Zheng et al. [8] show that many real-world legal queries have low lexical overlap with relevant documents and require multi-step inference, meaning high recall scores can mask retrieval for the wrong reasons.

The LRAGE framework [14] reinforces this through component-wise analysis, demonstrating that retrieval failures drive a disproportionate share of downstream hallucinations, and that generative performance cannot be meaningfully attributed without first isolating retrieval contributions. These findings motivate the multi-dimensional evaluation design adopted here, which assesses effectiveness, document-level reliability, and computational efficiency independently.

Across the reviewed literature, lexical retrieval remains competitive in structured statutory corpora, sparse neural methods such as SPLADE address vocabulary mismatch efficiently, and hybrid sparse–dense systems consistently outperform single-mode retrievers in regulatory and multilingual legal tasks.

Two specific gaps remain, however. First, no systematic evaluation of hybrid sparse–dense retrieval exists for EU climate-law statutes, which uniquely combine formulaic boilerplate with semantically complex regulatory obligations. Second, while multi-stage pipelines with reranking and LLM-based entailment verification have been proposed [2], [11], it

TABLE I
SUMMARY OF LITERATURE THEMES AND STUDY POSITIONING

Theme	Key Challenge	Proposed Solution	This Study
Retrieval Tension	Lexical precision vs semantic abstraction	Hybrid sparse-dense architectures [1], [3], [10]	Voyage-law-2 + SPLADE fusion
Reliability	Document-Level Retrieval Mismatch	SAC [12]	DRM reduction in EU corpus
Multi-turn Grounding	Context loss across turns	Query rewriting, LexRAG [7]	First-stage retrieval grounding
Terminology	Definition inconsistency	LexDrafter [9]	Terminological coherence
Authority Structure	Hierarchy ignored	Knowledge graphs, CBR-RAG [6], [13]	Doctrinal relevance vs precision
Efficiency	Multi-stage cost	Strong hybrid retriever [2], [11]	Retrieval-only evaluation

remains untested whether a strong first-stage hybrid retriever alone provides sufficient reliability, a question with direct practical implications for deployment cost. This study addresses both gaps by evaluating a Voyage-law-2 and SPLADE hybrid on EU climate-law documents across all three evaluation dimensions, providing the first targeted evidence for this domain.

VIII. CONCLUSION

This review has established that lexical and dense retrieval models address complementary failure modes in legal corpora, and that their combination is a principled response to the precision–abstraction tension consistently identified in prior legal IR research. It has further shown that effectiveness metrics alone are insufficient: document-level provenance [12], multi-turn grounding [7], terminological consistency [9], and doctrinal authority [6], [13] each constitute independent reliability requirements that standard Recall@k cannot capture. The LRAGE framework [14] and Zheng et al. [8] additionally demonstrate that retrieval quality is the primary determinant of downstream generation reliability, making first-stage retrieval design the most consequential architectural decision in legal RAG systems. The practicum that follows tests whether a hybrid Voyage-law-2 and SPLADE retriever can satisfy this multi-dimensional standard in the previously unevaluated domain of EU climate-law statutes, contributing targeted empirical evidence to an otherwise fragmented landscape.

IX. DISCLAIMER

Generative AI tools were used solely for proofreading and improving clarity of writing. All research ideas, technical content, and academic contributions are entirely our own.

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